

CHAPTER 1

Tang
&
Yang

Rong
Tang

Yun
Yang

2019–
2025

An-
nals
of
Sta-
tis-
tics
JRSS

MCMC

1.

1.1.

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$$D$$

$$\frac{d \ll D}{D}$$

 D GAN
VAE

(1)

(2)

(3)

MCMC

Tang-
Yang

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&
Yang
2019–
2025

MCMC

1

$\mathcal{Q}$
VAE

$\mathcal{Q}$

MCMC

2.

2.1.

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$$\mathcal{M}^*$$

$$\mathbb{R}^D$$

$$d$$

$$\beta$$
-

$$\beta \geq 1$$

$$P^*$$

$$\mathcal{M}^*$$

$$\mathrm{vol}_{\mathcal{M}^*}$$

$$p^*$$

$$\alpha-$$

$$\alpha \in [0,\beta-1]$$

$$p^*$$

$$\mathbb{R}^D$$

Lebesgue

Lebesgue

Par-
ti-
tion
of
Unity
Tang

$$(U_\lambda, \varphi_\lambda)$$

$$\{\gamma_\lambda\}$$

]]tang2022minimax

]]tang2023lowdim

Reach

reach

$$\text{reach}(\mathcal{M}) = \inf\{r > 0 : \mathcal{M}$$

r -

}

2.2.

IPM.

KL

Hellinger

Tang

IPM

$$d_{\mathcal{F}}(\mu, \nu) = \sup_{f \in \mathcal{F}} \left| \int_{\mathbb{R}^D} f(x) \, \mathrm{d}\mu - \int_{\mathbb{R}^D} f(x) \, \mathrm{d}\nu \right|.$$

$\mathcal{F}$ γ -
Hölder

Sobolev

1-
LipschitzWasserstein-
1

RKHS

MMD
 γ

trade-
off

$$\gamma = 1$$

$$d_\gamma$$

Wasserstein-
1

$$\gamma \rightarrow 0^+$$

$$d_\gamma$$

?]tang2025regression

2.3.

•
Tang

$$= n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}}.$$

$$n^{-1/2}$$

$$d$$

β -

d

Haus-
dorff

D

Lep-
ski

chain-
ing
peel-
ing
lo-
cal-
iza-
tion

Gir-
sanov

Poincaré

2.4.

**·
Gauss-
ian**

Langevin

$$p_{\sigma}(x) = \int p_{\text{data}}(y) \phi_{\sigma}(x - y) \, \mathrm{d}y$$

$$L_{\infty}$$

-

$$\nabla \log p_t(x)$$

$$t$$

—

$$t =$$

$$0$$

$$\frac{t}{T} =$$

?]tang2024diffusion

$$\mathcal{S}_{\text{NN}}$$

$$t$$

Girsanov

Gir-
sanov

SDE

3.

3.1. ?]]tang2022minimax

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and
Yun
Yang.

“Min-
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Losses.”
*An-
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tics*,
2022.

 D

GAN
VAE

γ

$$n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}},$$

 α β d

D

par-
ti-
tion
of
unity

VAE

Taylor

Lepski

Fano

Varshamov-Gilbert

]]tang2022minimax

]]tang2023lowdim,
]]tang2024diffusion

3.2. ?]]tang2023lowdim

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Yang.

“Estimating Distributions with Low-dimensional Structures Using Mixtures of Generative Models.”
ICML, 2023.

VAE
GAN

MLD-
MAE

K

-

ρ_k

1-
Wasserstein

α -

β -

d

$$O\Big(n^{-\frac{\alpha\wedge(\beta-1)+1}{2(\alpha\wedge(\beta-1))+d}}\Big)\vee O\Big(\frac{\log n}{\sqrt{n}}\Big),$$

$$\begin{array}{l} \alpha\leq \\ \beta- \\ 1 \end{array}$$

$$/$$

?]tang2022minimax

?]tang2022minimax

3.3. ?]]tang2023robust

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Rong
Tang,
Anir-
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Bhat-
tacharya,

Deb-
deep
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Yun
Yang.
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JRSS
B,
2023.

Stiefel

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Bayesian
RPE-
TEL

(1)

(2)

(3)

Bernstein-
von
Mises
(BvM)

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(RRWM)

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3.4. ?]]tang2023twosample

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Rong
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Yun
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Losses,”
*AIS-
TATS*,
2023.

MMD

$O(n^2)$

Besov

Besov

1 ℓ_1
 ℓ_2 **Wasser-
stein**

1

2

$$O(n \log n + n^{2d/(4\alpha+d)})$$

MMD

$$O(n^2)$$

**3.5. Tang
&
Yang
(2024a)**

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Rong
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TATS*,
2024.

SOTA

Langevin

—

d

D
Langevin

Gauss-
ian

$\tilde{O}(\sigma)$

σ

n

d

L_∞

—

1-
Wasserstein

$$\frac{1}{\sqrt{n}} \vee n^{-\frac{\alpha+1}{2\alpha+d}}.$$

?]tang2022minimax

?]tang2022minimax

3.6. [?\]\]tang2024mala](#)

Metropolis-
Adjusted
Langevin

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Rong
Tang
and
Yun
Yang.
“Metropolis-
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Di-
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De-
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and

Be-
yond.”
ICML,
2024.

MALA

MCMC

$$O(d^{1/3})$$

s-
conductance
pro-
file

MALA

$O(d^{1/3})$

**burn-
in**

s-
conductance
pro-
file

con-
duc-
tance

Chen
et
al.
(2020)

con-
duc-
tance
pro-
file

warm-
start

$\log M_0$

$\log \log M_0$

?]]tang2023robust

?]]tang2023robust

BvM

BvM

MALA

3.7. Tang
&
Yang
(2025a)

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Rong
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Yun
Yang.
“Min-
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Rates
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Dis-
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tions,”
arXiv,
2025.

X Y $\mu_{Y|X}^*$ X Y

IPM

(1) **Regime**
1

Lebesgue

$$n^{-\frac{\alpha_X}{2\alpha_X+d_X}}+n^{-\frac{\alpha_Y+\gamma}{2\alpha_Y+D_Y+\frac{\alpha_Y}{\alpha_X}\cdot d_X}}.$$

(2)

Regime

2

:

$\mathcal{M}_{Y|X}$

X

$$\begin{array}{l} n^{-\gamma\beta_Y/d_Y} \\ (3) \text{ Regime} \\ 3 \end{array}$$

:

$$\mathcal{M}_{Y|x}$$

$$x$$

$n^{-\gamma/(\frac{d_Y}{\beta_Y}+\frac{d_X}{\beta_X})}$

γ -
Hölder
IPM
 d_γ

$\gamma =$
0

1-
Wasserstein

$\gamma =$
1

3.8. Tang
&
Yang
(2025b)
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Rong
Tang
and
Yun
Yang.
“Con-
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are
Minimax-
Optimal
and
Manifold-
Adaptive,”
arXiv,
2025.

?]]tang2024diffusion

$$\alpha_X \in [0, 1]$$

(2)

X

Y

**Wasser-
stein**

(d_X, d_Y)

(D_X, D_Y)

lo-
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and
peel-
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tech-
niques

Gir-
sanov

?]]tang2024diffusion,
?]]tang2025regression

?]]tang2024diffusion

→
?]]tang2025regression

→

4.

Tang-
Yang

2019—
2025